The Fire Fighter Robot

Introduction

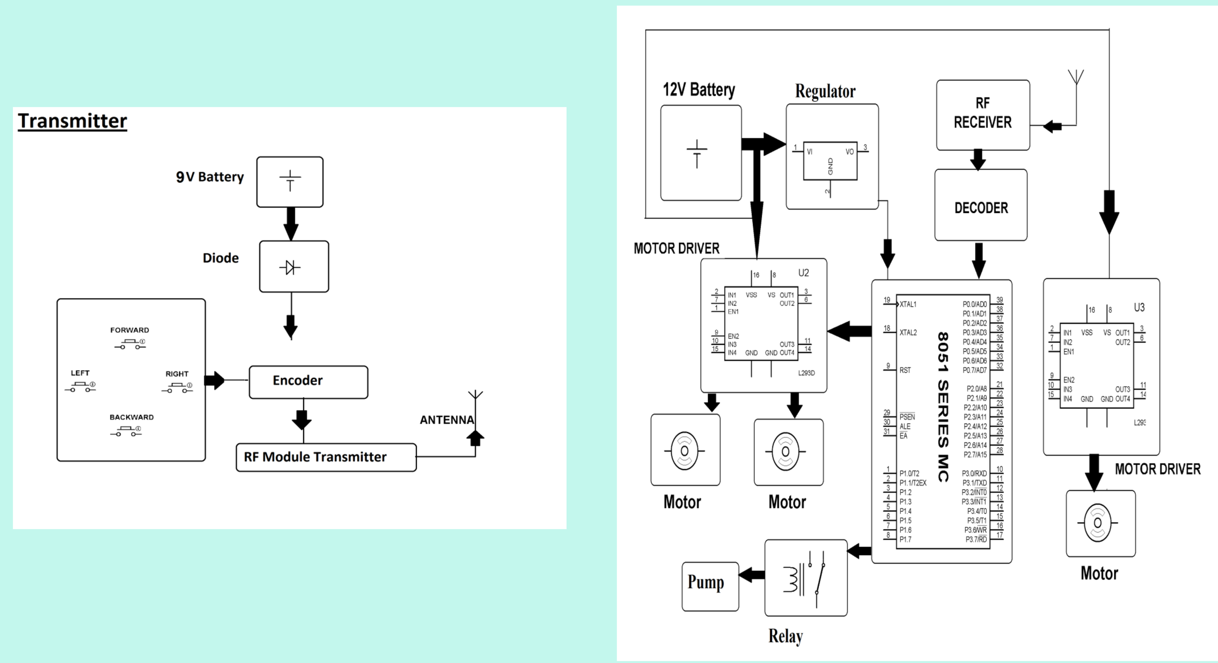
Fire incident is a disaster that can potentially cause the loss of life, property damage and permanent disability to the affected victim. Major fire accidents do occur in industries like nuclear power plants, petroleum refineries, gas tanks, chemical factories and other large-scale fire industries resulting in quite serious consequences. Thousands of people have lost their lives in such mishaps. Therefore, this project is enhanced to control fire through a robotic vehicle. With the advancement in the field of Robotics, human intervention is becoming less every day and robots are used widely for purpose of safety. In our day to day life fire accidents are very common and sometimes it becomes very difficult for fireman to save human life. In such case firefighting robot comes in picture.

Components

Hardware Specifications
Arduino UNO
Robotic Chassis
Water Tank
Spray Tube
Pump
bluetooth
Motor Driver
Gear Motor
IR sensor
Flame sensor
Bread board
Wheel
Cables and Connectors
Diodes
PCB and Breadboards
LED
Push Buttons
Switch
IC
Battary & Charger

Software Specifications
Keil µVision IDE
Arduino IDK
MC Programming
Language: C

Block Diagram



Application

- Can be used in extinguishing fire where probability of explosion is high. For eg. Hotel kitchens, LPG/CNG gas stores, etc.
- Can be used in Server rooms for immediate action in case of fire
- Every working environment requiring permanent operator's attention, At power plant control rooms
- Can be used in search and rescue operation
- Can Used in domestic cold storage places

Requirements

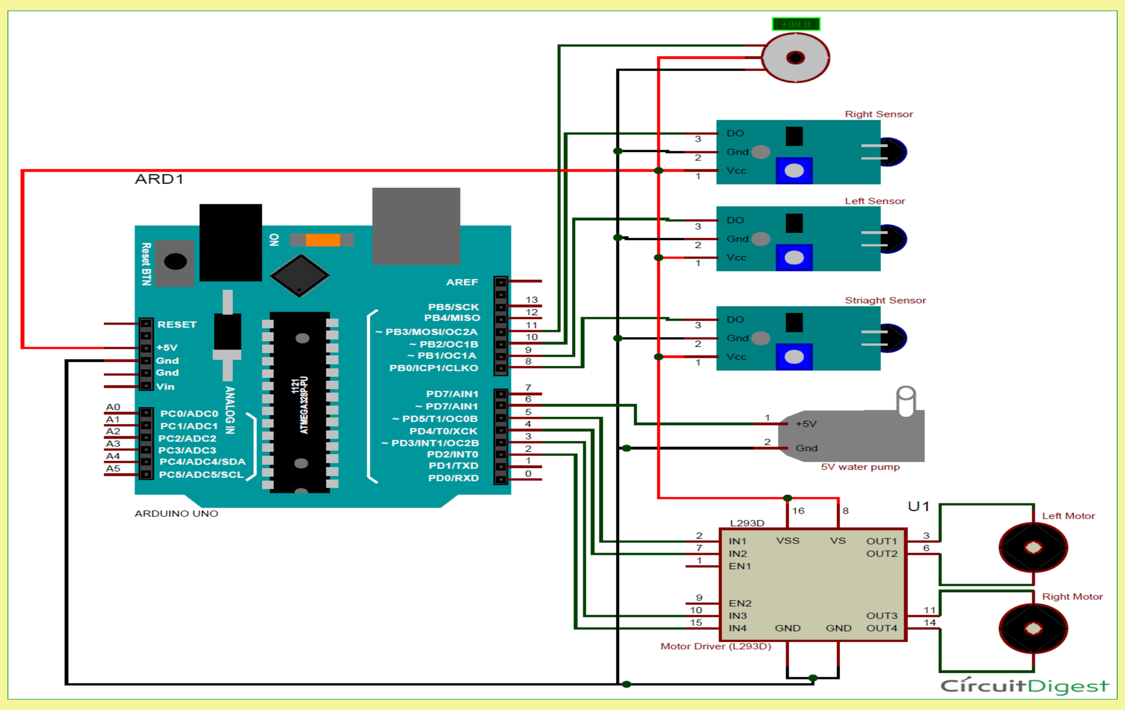
To design and manufacture a firefighting robot Specific objectives:

1. To design and implement a test environment
2. To design and implement an alarm system
3. To design and implement a fire detection system
4. To design and fabricate the frame of the robot
5. To design and implement a water dispensation system
6. To design and implement the drive system
7. To design and implement the control system
8. To design and Implement the temperature detection
9. To implement a camera sensor for path finding and object identification
10. To design and implement Artificial intelligence controlling system

Constrains:

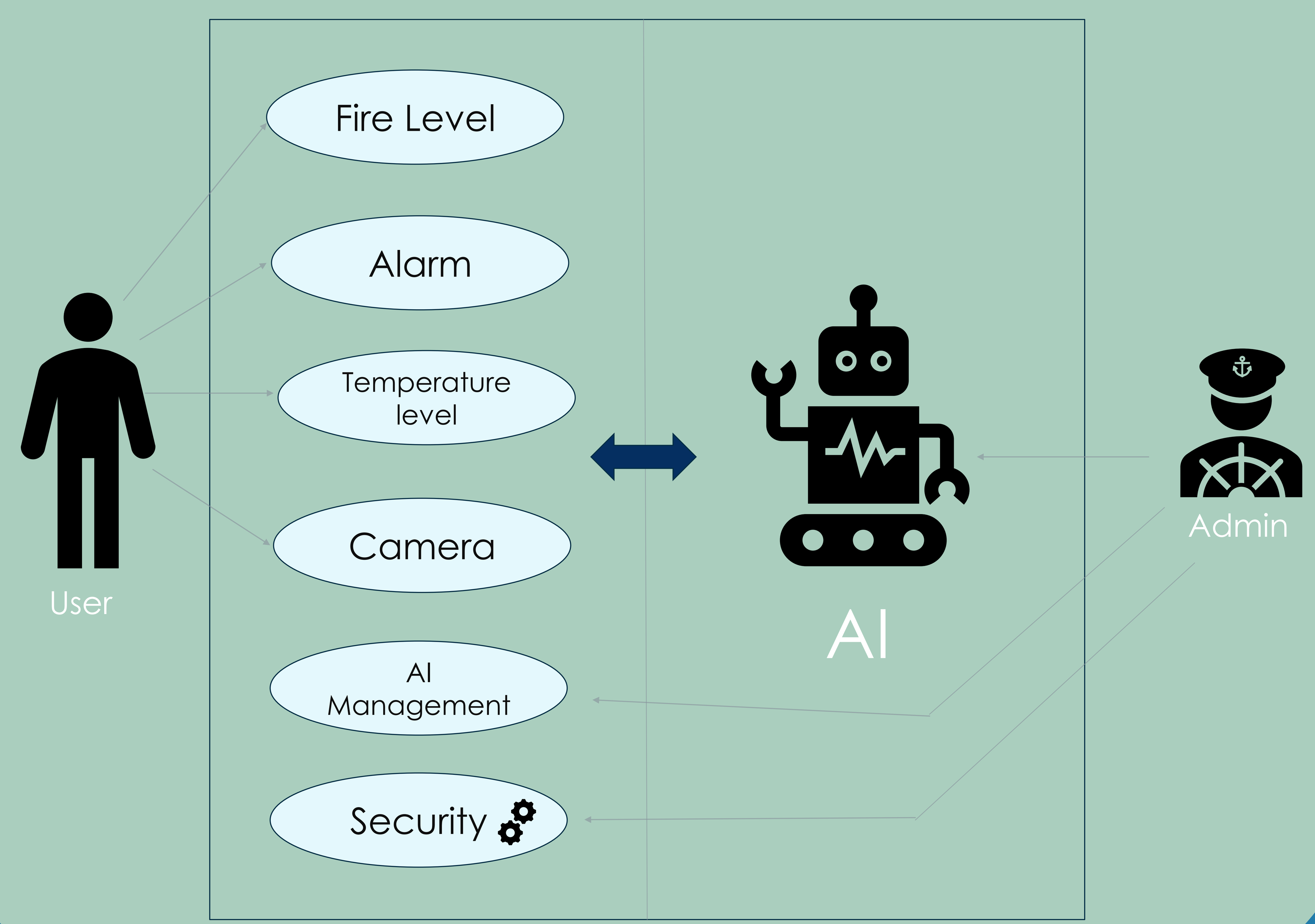
I. The robot cannot climb stairs
II. High skills required to implement some sections.
III. Some parts like the rubber wheels will melt at high temperature

Circuit Diagram



Use case Diagram

The fire fighter robot



Design and Prototype

Conclusion

The firefighting robot is a promising new technology that has the potential to revolutionize the way fire fighters operate. It is capable of navigating through a burning building and locating the source of the fire and extinguishing it quickly and accurately. Its main advantage is that it can be used in hazardous environments where it would be too dangerous for humans to enter. It also has the potential to save lives, as it is capable of responding to fires more quickly than human firefighters. Therefore, the firefighting robot is an excellent tool to have in the firefighting arsenal.

References:

Prof.Dr.S.N. Kini(An ISO 3297: 2007 Certified Organization) Vol. 4,Issue 11, November 2015 [10]
Swati Deshmukh, Karishma Matte and Rashmi Pandhare, "Wireless Fire Fighting Robot",
International journal for research in emerging science and technology, vol.-2, pp.373 - 375, March-2015. [11]
Adeel ur Rehman, Dan-Sorin Necsulescu, Jurek Sasiadek, "Robotic Based Fire Detection In Smart Manufacturing Facilities", IFAC.

Acknowledgement:

Sadiq Iqbql
Head, dept of CSE

Pobon Shaha
Lecturer, dept of CSE

MD Sobuj Ali
Mentor, Project Group

Project Showcase on
Intelligence perception